

Hoare Huang

黄浩然

+852 4681 8917

24101322d@connect.polyu.hk



CS undergraduate at **The Hong Kong Polytechnic University** (GPA 3.9 / 4.3, Sem 2 2025/26) working at the intersection of **AI engineering and finance**. Currently a Forward Deployed Engineer / AI Intern at **Magic Compass (HK)**, building an AI-Native risk-control panel. Shipped two independent production products in **three days** using AI-assisted development workflows. Cross-disciplinary reader in philosophy, history and the social sciences — treating rigorous systems engineering and thoughtful product design as one practice.

■ EXPERIENCE

Forward Deployed Engineer / AI Intern

Jun 3 – Aug 31, 2026

Magic Compass · Hong Kong

- Architected and shipped an **AI-Native risk-control panel** for a Hong Kong-based financial platform, automating internal workflows and replacing manual triage of platform events with policy-driven AI agents.
- Built **AI agents** that reason over real-time platform events — **withdrawal approvals, trading halts, mispricing detection** — producing human-readable, compliance-aware action recommendations for the operations team.
- Engineered the end-to-end pipeline from event ingestion through agent reasoning to operator-facing decisions, balancing AI judgment with financial-compliance constraints in a privacy-sensitive production environment.

■ PROJECTS

Independent Products

Copier — Windows Clipboard Island

Jul 2026 · 2 days

Founder & Sole Developer · nidecopier.com · v0.1.0 released

- 0 → 1 **designed, built and shipped** a Windows-native clipboard island tool, inspired by macOS clipboard managers.
- **Tech:** C# / .NET WPF (Windows), Swift (macOS port), Next.js (landing page, deployed on Vercel), SQLite (local-first storage); 3-layer architecture (App / Core / Interop).
- Global hotkey **Alt+V** summons a glass clipboard island with two morphs (follow-mouse / top-bar), search, pinning, and support for 6 content types (text / image / file / HTML / color / link); privacy-preserving by design — no keylogger, no cloud.
- **Shipped in 2 days** using AI-assisted development workflows; productized as free-10 + ¥20 one-time licence (no subscription).

OpenWisdom — Social-Science Agent Skills

Jul 2026 · 1 day

Founder & Sole Developer · openwisdom.vercel.app · GitHub · open source

- 0 → 1 **designed, built and open-sourced** a library of **Agent Skills** that package methods from **psychology, sociology, history, political science and economics** into AI Skills consumable by any modern coding agent.
- **Tech:** pnpm monorepo + TypeScript, packages schema / providers / catalog / core / cli / mcp / web (Next.js, deployed on Vercel); published to npm (openwisdom & openwisdom-mcp 0.1.0+).
- Dual install surface: a **CLI** (`npm openwisdom install`) and a **MCP stdio server** for in-session agent use; compatible with Claude Code, Cursor, Codex, Gemini CLI, GitHub Copilot and portable agents.
- **Shipped in 1 day** using AI-assisted development workflows.

■ PROJECTS

Coursework

Distributed Sensor Data Aggregation Platform

Feb – Apr 2026

PolyU · Core Developer (Rust)

- Designed and implemented a **Rust-based Buffer Manager** for a distributed data platform, allocating an independent thread per sensor to guarantee zero data loss and integrity under high concurrency.
- Achieved **100% data-collection accuracy under peak load** through buffer-capacity tuning and multi-thread coordination; no overflow or loss observed during system operation; project received a high score.

Multi-threaded TCP / HTTP Server

Mar – Apr 2026

PolyU · Computer Networks Course

- Implemented an RFC-compliant HTTP server supporting **persistent connections, conditional requests (304) and MIME type mapping** using Python's socket and threading modules.
- Hardened with path-traversal prevention (resolve() + prefix validation) and threading.Lock for concurrent log writes — **zero log loss at 500+ RPM**.
- Added Apache combined-format access logging, 5s idle timeout, and graceful shutdown; validated all edge cases via curl / Postman (100% request coverage).

■ EDUCATION

The Hong Kong Polytechnic University

GPA 3.9 / 4.3 · Sem 2, 2025/26

B.Sc. (Hons) in Computing

Core coursework: Operating Systems · Database Systems · Computer Networks · Data Structures & Algorithms · Software Engineering · Distributed Systems.

■ SKILLS

Languages

Rust Python TypeScript Java C C++ C# / .NET
Swift

Systems & Networking

TCP/IP · HTTP · Multi-threading · Memory Safety ·
Distributed Systems · Local-First Storage · Modular Design
· Concurrency & Lock Primitives

AI / Agent Harness

AI-Assisted Development · Agent **Skills & MCP** · Multi-Agent Orchestration · AI-Native Application Architecture · Prompt Engineering for Production

Domain

Financial-platform workflows (risk control, compliance, event automation) · Cross-disciplinary product thinking

■ CROSS-DISCIPLINARY INTERESTS

Reading: Philosophy, Psychology, History, Political Science, Sociology.

Translating long-form reading into operational frameworks — see **OpenWisdom**, where the methods become AI Skills consumable by any coding agent.

■ EXTRACURRICULAR

PolyU Mainland Students Football Team

Feb – Apr 2026

Starting Midfielder

- Coordinated tactical execution and in-game decision-making; led the team to the inter-university league **knockout stage** with **5 assists** in high-pressure matches.